

Dynamic Planet

Sierra Vista Invitational 2023

**Do not turn this page until your
Event Supervisor tells you to.**

Directions:

1. You are allowed to write on this test, but please do not write on the image sheet.
2. The test contains 69 questions for a total of 114 points. In general, the test begins with standalone questions and transitions to longer questions based on images or larger topics.
3. Image numbers reference the labeled images on the separate image sheet.
4. For calculations, only the final answer will be graded; partial credit will not be given. Partial credit may be given for explanations.
5. Some questions may ask you to select one or more choices. The number of points does not correlate to the number of correct answers.
6. Good luck!

1. Consider a hypothetical stream that is fed by 16 individual streams, each of which has no tributaries. (4 points)

Maximum possible Strahler stream order:

Minimum possible Strahler stream order:

Maximum possible Shreve stream order:

Minimum possible Shreve stream order:

2. Image 1 shows a cross-section of the topography in an area. Which of the following drainage patterns is LEAST likely to form here? (1 point)

- a. Dendritic
- b. Radial
- c. Rectangular
- d. Trellis
- e. Centripetal

3. Consider a comparison between a tree and a stream system. The roots of the tree uptake water which is then carried through the plant to its other parts. In this model, the roots can be considered tributaries/distributaries and the branches can be considered tributaries/distributaries. (1 point)

- a. Tributaries, tributaries
- b. Tributaries, distributaries
- c. Distributaries, tributaries
- d. Distributaries, distributaries

4. Which of the following types of lakes can exhibit internal waves during the winter? (1 point)

- a. Amictic
- b. Cold monomictic
- c. Warm monomictic
- d. Dimictic

5. Explain how algae blooms and extended periods of lake stratification can lead to similar conditions in a lake. (4 points)

6. The stratosphere is a stratified body of fluid that increases in temperature as you gain altitude. In what season does a dimictic lake most resemble these conditions? (1 point)
- Winter
 - Spring
 - Summer
 - Fall
7. Observing a lake, you notice that there are parallel rows of plant matter and bubbles running north to south. What direction could the winds blowing over this lake be coming from? (2 points)
8. What is the name of the process that creates the effects described in the previous question? (2 points)
9. Four distinct layers of water are in a lake with each layer having a uniform temperature and salinity. Layer 1 is 4.02 degrees celsius with salinity 0.01%; layer 2 is 4.02 degrees celsius with salinity 0.02%; layer 3 is 3.98 degrees celsius with salinity 0.02%; layer 4 is 3.98 degrees celsius with salinity 0.03%. Order the layers from lowermost to uppermost in the lake. (ex: 1, 4, 3, 2) (3 points)
10. What is the term for the localized reduction in groundwater level caused by water extraction at a specific point, such as a well? (1 point)
11. Images 2 and 3 show different trends of quantities with respect to distance from the stream mouth, i.e. the left side is CLOSER to the mouth. For the following quantities, select whether they are best represented by image 2 or 3: (4 points)
- Capacity:
- Competence:
- Stream velocity:
- Stream discharge:
- Stream gradient:
- Sinuosity:

Channel roughness:

Channel width:

12. Can you draw drainage divides for drainage systems that include a bifurcation lake?
Why? (2 points)

13. What is the name of the process in which two streams join together to form one stream?
(1 point)

14. What is the name of the process in which one stream splits into multiple streams?
(1 point)

15. A riffle pool sequence describes rhythmic changes in what property of a stream?
(2 points)

- a. Bank width
- b. Water speed
- c. Sinuosity
- d. Stream roughness coefficient
- e. Stream depth

16. Which of the following streams would most likely exhibit riffle-pool sequences?
(1 point)

- a. Straight with competence in the gravel ranges
- b. Straight with competence in the silt ranges
- c. Meandering with competence in the gravel ranges
- d. Meandering with competence in the silt ranges

17. Which of the following would NOT result in the creation of a knickpoint? (1 point)

- a. A stream's substrate suddenly changing from limestone to granite
- b. A stream flowing over a normal fault from the footwall to the hanging wall
- c. A stream flowing over a reverse fault from hanging wall side to footwall side
- d. A glacier forming a hanging valley and then melting

18. Take a look at Image 4. Using what you know about rock types, which of the following is most likely the rock that these potholes are formed from? (1 point)
- a. Granite
 - b. Gabbro
 - c. Schist
 - d. Limestone
19. What are the units of drainage density? (1 point)
- a. length
 - b. 1/length
 - c. $\text{length}^2/\text{mass}$
 - d. length/mass
20. Suppose the time of concentration for watershed A is 2 hours while for watershed B it is 1 hour. Which watershed would have higher peak discharge at the outlet after the exact same rainstorm occurs over each watershed? Which watershed would have a sharper rising limb on a hydrograph? Which watershed would return to its base flow level faster? (Assume that all rain enters the watershed evenly.) (1 point)
- a. A, A, A
 - b. B, A, B
 - c. A, A, B
 - d. B, B, B
21. A sample of water from a nearby aquifer is compared to one taken of fossil water from another aquifer. This fossil water infiltrated the ground back during the Carboniferous period which was dominated by the presence of trees. Which would you expect to have the higher dissolved oxygen? Why? (3 points)
22. Which of the following is NOT a major direct contributor to freshwater acidification? (1 point)
- a. Leaching of soil and rock
 - b. Dissolution of carbon dioxide into water
 - c. Reduction of sulfur oxides and nitrogen oxides
 - d. Acid rain

23. An example of a negative feedback loop is in the production of CO₂, increasing strength of the greenhouse effect, and dissolution of CO₂ in freshwater. Due to this, would you expect the solubility of CO₂ in water to increase or decrease given a decreased temperature? (0.5 points)
- a. Increase
 - b. Decrease
24. Three water drop penetration time tests are done on Three different soils to measure the hydrophobicity of the samples. They ended up finding that none of the samples were particularly hydrophobic, but they did learn about the soil's composition. In sample 1, the soil was a vast majority clay; in sample 2, the soil was mostly sand; and in sample 3, the soil was mostly loam. The times measured are 15 seconds, 10 seconds, and 5 seconds. What sample did each measurement come from? (3 points)
- 15 seconds:
- 10 seconds:
- 5 seconds:
25. The hyporheic zone and the riparian zone are very similar in that they are both interfaces between a stream and are important for regulating chemicals and nutrients that are present in a stream. The hyporheic zone is the boundary zone between a stream and ____, while the riparian zone is the boundary zone between a stream and _____. (2 points)
26. Which of the following freshwater systems are lotic? Select all that apply. (1 point)
- a. Lakes
 - b. Swamps
 - c. Streams
 - d. Mires
 - e. Hyporheic zone
27. When considering all freshwater on Earth, do lentic or lotic systems store more water? (0.5 points)
- a. Lentic
 - b. Lotic
28. What is the name of the process shown in Image 5? (1 point)

29. What type of weathering does the process in Image 5 show? (1 point)
- a. Biological
 - b. Chemical
 - c. Mechanical
30. It turns out that yazoo streams were named after an actual stream/river that exhibits what we now know as yazoo behavior. What major river did the original Yazoo River branch off from? (1 point)
- a. Mississippi
 - b. Nile
 - c. Yangtze
 - d. Mekong
31. Identify the structure shown in Image 6. (1 points)
32. Artesian wells form when what “surface”, the line to which groundwater rises under hydrostatic pressure, is located above ground level? (1 point)
33. The Amazon river discharges about 4.381 million cubic feet per second. Suppose an average first-order stream discharges 10 cubic meters per second. Assuming no water evaporates while flowing, what would be the Shreve order of the Amazon river? (1 meter = 3.28 feet) (2 points)
34. Image 7 is a picture of a stream at Umstead State Park in Raleigh, North Carolina. What is the name for the feature labeled “A” where sediment is being deposited? (2 points)

35. Is the stream pictured in Image 7 generally low or high energy? Provide two pieces of evidence that support your conclusion. (3 points)

36. Image 8 shows some nice frozen dirt, also from Umstead in North Carolina. The dirt is most definitely frozen (I poked it). Even though this is frozen ground, why is it not considered permafrost? (2 points)

The next 3 questions refer to Image 9, three pictures of shirts at Uniqlo. Suppose that the seams are cross sections of river valleys. (Note the extra triangular seams in picture A.)

37. Which of the following types of rivers would be found in a river valley that looks like shirt A? (1 point)

- a. Youthful river
- b. Mature river
- c. Rejuvenated river

38. Which of the following types of rivers would be found in a river valley that looks like shirt B? (1 point)

- a. Youthful river
- b. Mature river
- c. Rejuvenated river

39. Which of the following types of rivers would be found in a river valley that looks like shirt C? (1 point)

- a. Youthful river
- b. Mature river
- c. Rejuvenated river

Suppose you are the god of rain (and other precipitation). Answer the following questions on your (scientifically accurate) powers:

40. In order for clouds (the all-important things that give you power) to form, small particles need to be present in the air. What term describes these particles? (1 point)
41. Which of the following could be a source for these small particles? (2 points)
- a. Volcanic outgassing
 - b. Evaporation of seawater
 - c. Cellular respiration
 - d. Transpiration in plants
42. When the previously mentioned small particles are present in the air, water condenses onto them forming ice crystals. What is the name of this process? (1 point)
43. Image 10 shows the different types of precipitation based on the temperature of the air it falls through. Label the types of precipitation from A-D. (4 points)
- A:
- B:
- C:
- D:
44. You now decide to send forth rain from your holy cloud castle. What shape do the raindrops make as they fall? (1 point)
45. On its way to the ground, your raindrop may react with what molecule, which can create a weak acid? (1 point)
46. How many oxygen atoms are involved in the reaction in the previous question? (1 point)

47. Your raindrop landed in Florida, which has bedrock mainly formed from what group of rocks? (1 point)

Image 11 shows a profile of a hypothetical stream. The bottom of the image represents sea level.

48. What type of point is point A? (1 point)

- a. Critical point
- b. Divide
- c. Nickpoint
- d. Saddle point

49. Suppose the stream is not constant, but only occasionally flows upstream of A. What feature would be formed at A? (1 point)

- a. Alluvial fan
- b. Cataract
- c. Delta
- d. Playa

50. Notice how the stream slowly approaches the black line. What does this line represent? (1 point)

51. T/F: The stream can never go below the black line. (0.5 points)

- a. True
- b. False

52. Briefly explain why the black line is not at sea level. What does this indicate about the stream? (2 points)

53. What does the white line represent? (1 point)

54. T/F: The white and black lines converge at the mouth of the stream. (0.5 points)

- a. True
- b. False

55. T/F: The white line can never go below the black line. (0.5 points)

- a. True
- b. False

56. Estimate the maximum stream gradient outside of A in ft/ft. (1 mi = 5280 ft) (2 points)

57. Estimate the maximum hydraulic gradient of the groundwater beneath the stream in ft/ft.
(2 points)

A month ago, California was hit by a series of large atmospheric rivers. These resulted in massive storms and flooding.

58. The storms produced a large amount of precipitation in a very short time. Considering California's drought conditions, would this downpour be better for the drought than the same amount of rain occurring over a long period of time? Why or why not? (3 points)

59. Now considering similar conditions as the previous question, would precipitation over the Sierra Nevada mountains or over California's Central Valley be more beneficial to the drought conditions there? Why? (Hint: These storms occurred in winter.) (3 points)

60. Because of these storms, the ground has become saturated. Which of the following statements are true regarding a new hydrograph for storms of the same size, before and after major rainfall events? Select all that apply. (2 points)

- a. Steeper rising limb
- b. More gradual rising limb
- c. Longer lag time
- d. Shorter lag time
- e. Lower peak
- f. Higher peak
- g. Higher baseflow
- h. Lower baseflow

61. Which of the following would increase the severity of flooding? Select all that apply. (2 points)

- a. Deforestation
- b. Plugging up crevasse splays
- c. Urbanization
- d. Creation of percolation trenches
- e. Degradation of riparian zone

62. Which of the following freshwater features typically occurs at the water table? Select all that apply. (2 points)

- a. Base level of a stream well above sea level
- b. Boundary between the acrotelm and catotelm of an undisturbed peat bog
- c. Stream surface level of a gaining stream
- d. Hyporheic zone of a stream
- e. Zone of dispersion during saltwater intrusion

63. Order the following layers of water in terms of increasing average thickness: 1) pellicular/bound water; 2) Capillary fringe; 3) Epilimnion. (2 points)

64. Identify the structures present at each of the labeled points in Image 12: (4 points)

A:

B:

C:

D:

65. What kind of rock is most commonly found in regions with karst topography? What is the chemical formula of its dominant mineral? (2 points)

Images 13 and 14 show Lake Baikal in Russia and Crater Lake in Oregon, respectively.

66. Which of these lakes has a higher shoreline development index? (0.5 points)

- a. Lake Baikal
- b. Crater Lake

67. Which of these lakes likely has the most nutrients? (0.5 points)

- a. Lake Baikal
- b. Crater Lake

68. Which of these lakes likely has a higher rate of evaporation? (0.5 points)

- a. Lake Baikal
- b. Crater Lake

69. Describe how each of these lakes likely formed based on their shape and other clues in the map. (4 points)

Lake Baikal:

Crater Lake: